Actividades de subneteo

Motivacion

Imaginemos que toda una universidad tuviera una sola red para:

💻 laboratorios
📶 WiFi
🖨 impresoras
📹 cámaras
🗄 servidores
🧑‍🏫 docentes
🧑‍🎓 estudiantes

Todo mezclado en una sola red.
¿Creen que funcionaría de manera eficiente?

Hoy descubriremos cómo los ingenieros de redes organizan y dividen grandes redes utilizando subneteo FLSM para mejorar:

✅ rendimiento
✅ organización
✅ seguridad
✅ comunicación

“Las redes inteligentes no crecen desordenadas… se diseñan.”

Saberes previos

Antes de iniciar la clase, reflexionemos:
✅ ¿Qué es una dirección IPv4?
✅ ¿Qué función cumple la máscara de subred?
✅ ¿Qué diferencia existe entre dirección de red y broadcast?
✅ ¿Qué ocurriría si todos los dispositivos estuvieran en una sola red gigante?
✅ ¿Qué entiendes por subneteo?
Hoy conectaremos todos estos conocimientos para aprender a crear subredes reales como lo hacen los ingenieros Cisco.

Logro dela sesion

- Al finalizar la sesión, el estudiante será capaz de aplicar subneteo IPv4 mediante FLSM para dividir una red en subredes organizadas, calculando correctamente:
  dirección de red
  dirección broadcast
  primera y última dirección IP útil
  cantidad de hosts y subredes
- Asimismo, implementará escenarios prácticos utilizando configuraciones básicas en entornos Cisco Packet Tracer.

Network Segmentation

Reasons for Subnetting

Large networks need to be segmented into smaller sub-networks, creating smaller groups of devices and services in order to:

- Control traffic by containing broadcast traffic within subnetwork
- Reduce overall network traffic and improve network performance

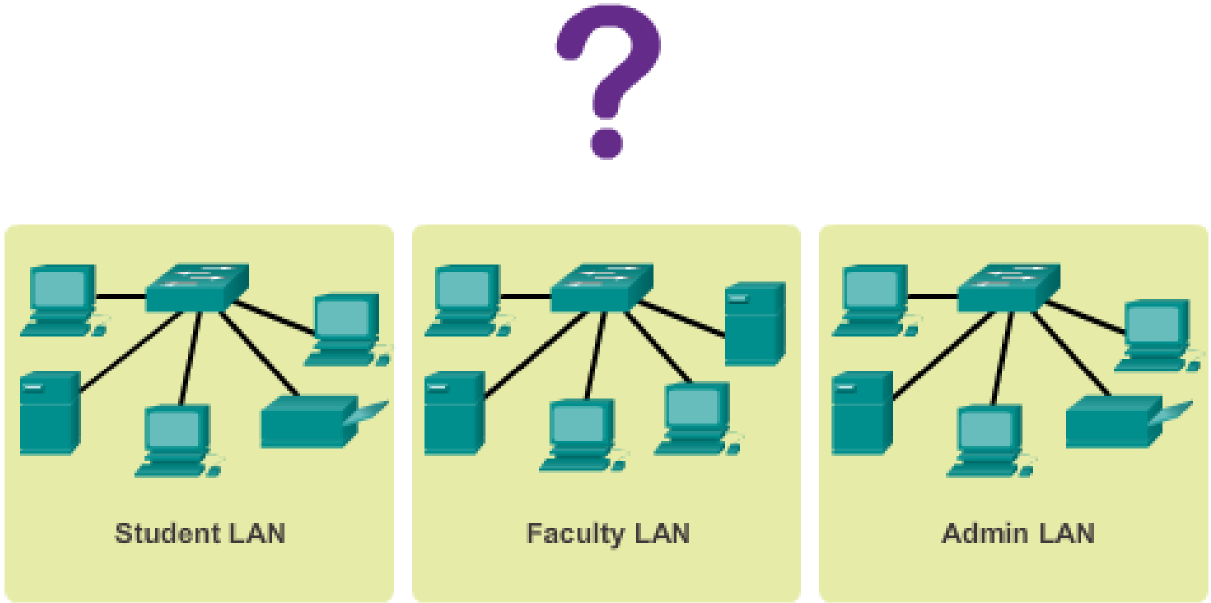
Subnetting - process of segmenting a network into multiple smaller network spaces called subnetworks or **Subnets.**

Communication Between Subnets

- A router is necessary for devices on different networks and subnets to communicate.
- Each router interface must have an IPv4 host address that belongs to the network or subnet that the router interface is connected to.
- Devices on a network and subnet use the router interface attached to their LAN as their default gateway.

Subnetting an IPv4 Network

IP Subnetting is FUNdamental



Planning requires decisions on each subnet in terms of size, the number of hosts per subnet, and how host addresses will be assigned.

Basic Subnetting

- Borrowing Bits to Create Subnets
- Borrowing 1 bit $2^1 = 2$ subnets

Address	192	168	1	0000 0000
Mask	255	255	255	0000 0000
	Network Portion			Host Portion

Original	192.	168.	1. 0	000 0000	Network 192.168.1.0/24
Mask	255.	255.	255. 0	000 0000	Mask: 255.255.255.0

Borrowing 1 Bit from the host portion creates 2 subnets with the same subnet mask

Subnet 0

Network 192.168.1.0-127/**25**

Mask: 255.255.255.**128**

Subnet 1

Network 192.168.1.128-255/**25**

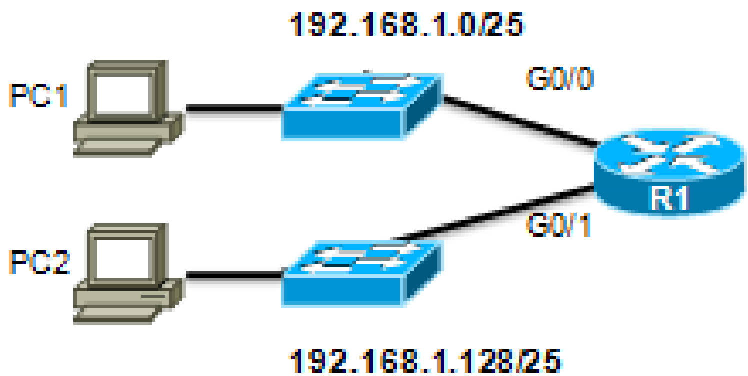
Mask: 255.255.255.**128**

Subnetting an IPv4 Network

Subnets in Use

Subnet 0

Network 192.168.1.0-127/25



Subnet 1

Network 192.168.1.128-255/25

Address Range for 192.168.1.0/25 Subnet

Network Address	192. 168. 1. 0 000 0000	= 192.168.1.0
First Host Address	192. 168. 1. 0 000 0001	= 192.168.1.1
Last Host Address	192. 168. 1. 0 111 1110	= 192.168.1.126
Broadcast Address	192. 168. 1. 0 111 1111	= 192.168.1.127

Address Range for 192.168.1.128/25 Subnet

Network Address	192. 168. 1. 1 000 0000	= 192.168.1.128
First Host Address	192. 168. 1. 1 000 0001	= 192.168.1.129
Last Host Address	192. 168. 1. 1 111 1110	= 192.168.1.254
Broadcast Address	192. 168. 1. 1 111 1111	= 192.168.1.255

Subnetting Formulas

- Calculate Number of Subnets

Subnets = 2^n
(where n = bits borrowed)

192. 168. 1. 0 000 0000

1 bit was borrowed

2^1 = 2 subnets

- Calculate Number of Hosts

Hosts = 2^n
(where n = host bits remaining)

192. 168. 1. 0 000 0000

7 bits remain in host field

2^7 = 128 hosts per subnet

Creating 4 Subnets

- Borrowing 2 bits to create 4 subnets. **2^2 = 4 subnets**

Borrowing 2 Bits

Original	192.	168.	1.	00	00 0000
Mask	255.	255.	255.	00	00 0000

Borrowing 2 bits creates 4 subnets:

Net 0	192.	168.	1.	00	00 0000	192.168.1.0/26
Net 1	192.	168.	1.	01	00 0000	192.168.1.64/26
Net 2	192.	168.	1.	10	00 0000	192.168.1.128/26
Net 3	192.	168.	1.	11	00 0000	192.168.192/26

All 4 subnets use the same mask:

Mask	255.	255.	255.	11	00 0000	Mask: 255.255.255.192

Subnetting an IPv4 Network

Creating 8 Subnets

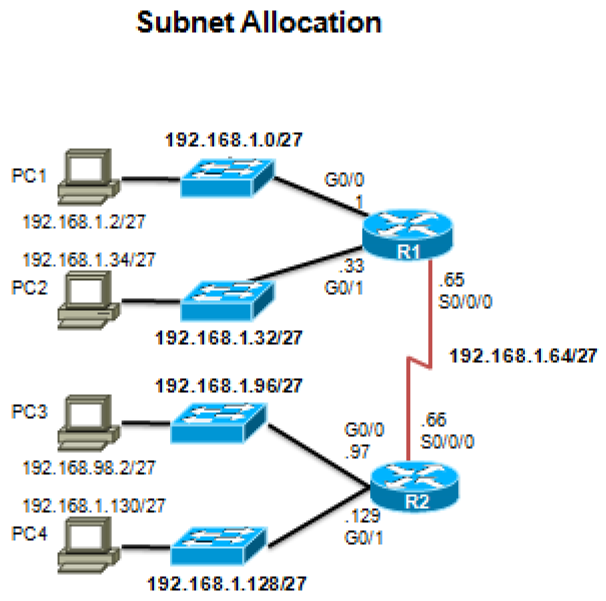
- Borrowing 3 bits to Create 8 Subnets. **2^3 = 8 subnets**

Net 0	Network	192.	168.	1.	000	0	0000	192.168.1.1
	Fist	192.	168.	1.	000	0	0001	192.168.1.1
	Last	192.	168.	1.	000	1	1110	192.168.1.30
	Broadcast	192.	168.	1.	000	1	1111	192.168.1.31
Net 1	Network	192.	168.	1.	001	0	0000	192.168.1.32
	Fist	192.	168.	1.	001	0	0001	192.168.1.33
	Last	192.	168.	1.	001	1	1110	192.168.1.62
	Broadcast	192.	168.	1.	001	1	1111	192.168.1.63
Net 2	Network	192.	168.	1.	010	0	0000	192.168.1.64
	Fist	192.	168.	1.	010	0	0001	192.168.1.65
	Last	192.	168.	1.	010	1	1110	192.168.1.94
	Broadcast	192.	168.	1.	010	1	1111	192.168.1.95
Net 3	Network	192.	168.	1.	010	0	0000	192.168.1.96
	Fist	192.	168.	1.	010	0	0001	192.168.1.97
	Last	192.	168.	1.	010	1	1110	192.168.1.126
	Broadcast	192.	168.	1.	010	1	1111	192.168.1.127

Subnetting an IPv4 Network

Creating 8 Subnets(continued)

Net 4	Network	192.	168.	1.	100	0 0000	192.168.1.128
	Fist	192.	168.	1.	100	0 0001	192.168.1.129
	Last	192.	168.	1.	100	1 1110	192.168.1.158
	Broadcast	192.	168.	1.	100	1 1111	192.168.1.159
Net 5	Network	192.	168.	1.	101	0 0000	192.168.1.160
	Fist	192.	168.	1.	101	0 0001	192.168.1.161
	Last	192.	168.	1.	101	1 1110	192.168.1.190
	Broadcast	192.	168.	1.	101	1 1111	192.168.1.191
Net 6	Network	192.	168.	1.	110	0 0000	192.168.1.192
	Fist	192.	168.	1.	110	0 0001	192.168.1.193
	Last	192.	168.	1.	110	1 1110	192.168.1.222
	Broadcast	192.	168.	1.	110	1 1111	192.168.1.223
Net 7	Network	192.	168.	1.	111	0 0000	192.168.1.224
	Fist	192.	168.	1.	111	0 0001	192.168.1.225
	Last	192.	168.	1.	111	1 1110	192.168.1.254
	Broadcast	192.	168.	1.	111	1 1111	192.168.1.255



Determining the Subnet Mask

Subnetting Based on Host Requirements

There are two considerations when planning subnets:

- Number of Subnets required
- Number of Host addresses required
- Formula to determine number of useable hosts

2^n-2

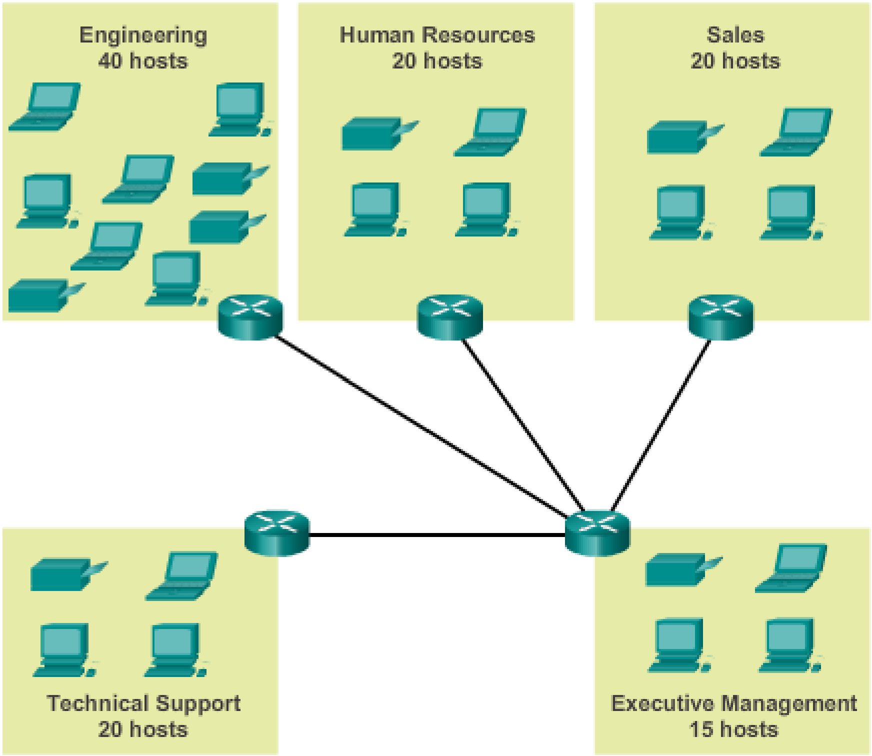
2^n (where n is the number the number of host bits remaining) is used to calculate the number of hosts

-2 Subnetwork ID and broadcast address cannot be used on each subnet

Determining the Subnet Mask

Subnetting To Meet Network Requirements

- It is important to balance the number of subnets needed and the number of hosts required for the largest subnet.
- Design the addressing scheme to accommodate the maximum number of hosts for each subnet.
- Allow for growth in each subnet.



Determining the Subnet Mask

Subnetting To Meet Network Requirements (cont)

Subnets and Addresses

	10101100.00010000.00000000.00000000	172.16.0.0/22
0	10101100.00010000.00000000.00000000	172.16.0.0/26
1	10101100.00010000.00000000.01000000	172.16.0.64/26
2	10101100.00010000.00000000.10000000	172.16.0.128/26
3	10101100.00010000.00000000.11000000	172.16.0.192/26
4	10101100.00010000.00000001.00000000	172.16.1.0/26
5	10101100.00010000.00000001.01000000	172.16.1.64/26
6	10101100.00010000.00000001.10000000	172.16.1.128/26
	Nets 7 – 14 not shown	
15	10101100.00010000.00000011.10000000	172.16.3.128/26
16	10101100.00010000.00000011.11000000	172.16.3.192/26

2^4 = 16 subnets

2^6 – 2 = 62 Hosts per subnet